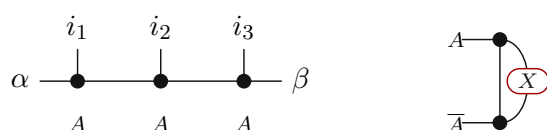


The **tenkz** language manual

a compact contract for tensor-network diagrams

The TNLean project — July 2026



A picture chooses one layout; its options declare topology and policy; its body declares atoms, connections, regions, and annotations.

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1 Install and read the language

The complete installation is two lines:

```
\usepackage{tenkz}
\begin{tenkz}[physical=up] \tn{A} & \tn{B} \end{tenkz}
```

For a repository checkout, put `tex/tenkz/` on `TEXINPUTS`; for an archived article, ship `tenkz.sty` and its package files beside the source. XeLaTeX is the reference engine for print and for the XDV-to-SVG web route.

The language contract is one sentence:

A picture chooses one layout; its options declare topology and policy; its body declares atoms, connections, regions, and annotations.

Read a source in that order. The environment chooses the placement grammar. Its optional argument states facts shared by the picture. Body commands declare records; they do not perform an imperative drawing procedure.

Five classes cover the public vocabulary: *atoms*; *connections and closures*; *regions and annotations*; *composition*; and *setup and extension*. A command declares an object in one of those classes. A key modifies the record being declared. This distinction keeps the grammar small enough to read from source.

Scope is explicit. `\tnset` is document scope; an environment option is picture scope; an atom option is object scope; and a join or side-policy option is connection scope. A shared key name has one value type and one meaning in every environment that exposes it.

Open ink is mathematical data. A matrix keeps its two virtual indices; a map keeps input and output indices; a scalar has none.

Closing a picture for neatness changes the object. The versioned `.tnlog` records the resolved model and boundary signature used for rendering. It also records every port's type, which the page does not: a contracted index carries its type in the record and not in its line, so a physical contraction and a virtual one stroke alike.

2 Which environment?

Choose from topology, not from the desired appearance.

Layout	Use it when	Recognition test
<code>tenkz</code>	The picture is a regular one-dimensional contraction.	Rows are layers, columns are sites, and neighbouring cells contract.

A spatial lattice, sheet, window, or measured region is not a second layout: it is a `tenkz` picture with a frame – `lattice=` or `planes` sugar, or a declared basis.

Do not change layouts for style. The layout shares the metric, atom skins, semantic roles, labels, and connection policy across every picture. A chain does not become a coordinate drawing because it is vertical. Choose the layout once, then declare policy through its public keys.

When none fits. State the missing mathematical class before adding syntax. A new key needs three manifested RMP consumers. A new command needs a new grammatical class. Until that gate passes, the honest result is a capability request, not raw TikZ hidden inside a canonical source.

3 A twenty-minute tutorial

Everything you will type here is the kernel surface: the one language the package binds the moment it loads. If you have met an older `tenkz` document, two sentences make you current. `\tenkzkernel` is an inert word — it rebinds what loading already bound — and the spellings it once switched on are dead, tombstoned in Appendix A. New sources need neither.

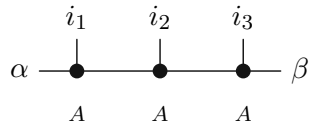
Eight examples take you from one word of tensors to the expectation value of a two-dimensional state, and each adds exactly one idea to the one before it. The objects are the ones the tensor-network literature draws — the matrix product state, its transfer matrix, the gauge identity, the PEPS and its double layer — and each is named in that language as it arrives. Every source is complete: paste it into any document that loads the package and it compiles as printed. Read each one in the same order — layout, picture policy, then body declarations. This tutorial needs only `tenkz` and, at the equation step, `tenkzeq`; when a figure of your

own wants a ring or another carrier this ladder does not climb, the chooser in Section 2 decides by topology.

3.1 A word of tensors

The first object is the matrix product state: one tensor A per site, with two virtual indices of bond dimension D that neighbouring sites sum over and one physical index of dimension d that stays free. Here are three sites of it.

EXAMPLE 3.1 · three sites of a matrix product state¹



```
% Formula: (A^{i_1} A^{i_2} A^{i_3})_{\alpha\beta}
% Ink: bonds contract; typed ports declare the open indices.
\begin{tenkz}[cols=3]
  \tn[ports={180:virtual:$\alpha$, 0:virtual, 90:physical:$i_1$}]{A} &
  \tn[ports={180:virtual, 0:virtual, 90:physical:$i_2$}]{A} &
  \tn[ports={180:virtual, 0:virtual:$\beta$, 90:physical:$i_3$}]{A}
\end{tenkz}
```

$$(A^{i_1} A^{i_2} A^{i_3})_{\alpha\beta}$$

¹ The horizontal bonds are summed virtual indices. The two stubs are the open matrix indices α and β ; each upward leg is one physical index.

kernel-boundary|signature=open:e,
open:w, phys:n, phys:n, phys:n

You have drawn the matrix-product word $(A^{i_1} A^{i_2} A^{i_3})_{\alpha\beta}$. **tenkz** holds one typed tensor network in one frame: **cols=3** lays three cells on a wire, each **\tn** declares an atom in the next cell, and **&** advances one site. Adjacent cells contract by default — those are the virtual bonds — so the only indices you declare are the open ones. **ports=** types each face and may label it; a port is the point where an index line meets its tensor. A typed port with no partner is an open index — an uncontracted one. Open ink is declared content: the picture exposes exactly the indices its atoms state, and the margin quotes the boundary signature — the list of the picture’s uncontracted indices — that the compiler wrote to the **.tnlog** event stream: two virtual stubs, three physical legs, nothing you did not say.

Beamer reads a frame’s body before typesetting it, so a chained picture reaches the environment with its **&** already an ordinary alignment tab. The picture reads its body as captured tokens and finds the tab there itself, so it compiles in a plain frame; no **[fragile]** marking is needed.

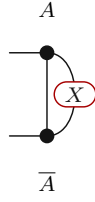


EXERCISE 3.1 Draw the plain matrix M — one atom, both virtual indices open, no physical leg — and say what boundary signature, what list of uncontracted indices, it writes.

3.2 Layers and side policy

One new idea: rows are layers. The object is the map every matrix-product calculation applies and reapplies, the transfer channel $\mathcal{E}_A(X) = \sum_i A^i X A^{i\dagger}$ — a ket layer over a bra layer, with the matrix X closing the input side.

EXAMPLE 3.2 · the channel applied to X ²



```
% Formula: E_A(X) = sum_i A^i X A^{i dagger}.
% Boundary: the input closes east; the matrix output stays
% open west.
\begin{tenkz}[rows={ket,bra}, cols=1, west=open, east={cup=$X$}]
  \tn[label pos=90]{A} \
  \tn[label pos=270]{$\overline{A}$}
\end{tenkz}
```

rows= declares a ket layer over a bra layer, and the frame contracts their facing physical ports — the vertical bond that sums i . The conjugate is label mathematics: $\overline{A}$ is a name, not a flag. The sides then state policy from a four-word alphabet. **open** mints legs and puts them in the signature; **none**, the default, is a side with no indices at all; **trace** returns a row to itself; **cup** closes adjacent layers onto each other. Here **east={cup=\$X\$}** closes the pair through the matrix X , and the west pair stays open, so the picture is a map on virtual matrices — which is exactly what its two-entry signature says.

The labelled cup is sugar: it expands to the side word **cup** plus a ring atom standing halfway along the generated cup wire, so the kernel record stream never contains a “labelled cup”, only a wire named **cup-1-2** and an atom on it. Every sugar row in the registry states its expansion this way, and `\tnset{strict}` rejects sugar outright.



EXERCISE 3.2 Draw the transfer matrix $E = \sum_i A^i \otimes \overline{A}^i$ of the state — the map on the D^2 -dimensional doubled virtual space: ket over bra, the physical pair contracted, all four virtual ends open.

EXERCISE 3.3 A contraction loop with no tensor on it — the spelling `\tnwire[closed]{(1,1)}{(2,1)}` — refuses to compile. Why?

3.3 The boundary alphabet, three ways

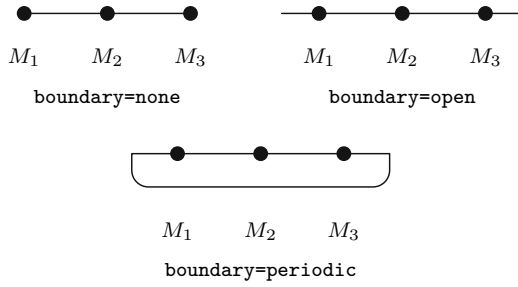
One row of matrices is three objects of the literature: a word with trivial boundary, the open-boundary matrix product $(M_1 M_2 M_3)_{\alpha\beta}$, and the periodic trace $\text{tr}(M_1 M_2 M_3)$ — the coefficient of a matrix product state with open and with periodic boundary conditions. The pair spelling **boundary=** states **west=** and **east=** at once, and its three words draw all three objects from one row:

$$\mathcal{E}_A(X) = \sum_i A^i X A^{i\dagger}$$

² X closes the east input pair. The west pair remains open because the output is a matrix; the vertical connection sums the physical index i .

kernel-boundary|signature=open:w, open:w

EXAMPLE 3.3 · one row, three boundary conditions³



```
% Formula: M_1 M_2 M_3 under three side policies.
\begin{tenkz}[rows={wire}, cols=3, variant]
  \tn{$M_1$} & \tn{$M_2$} & \tn{$M_3$}
\end{tenkz}
```

One family, one figure: the three panels share every token except the word under them. **none** is the explicit spelling of the default — a side that says nothing exposes nothing, the way a bond that closes trivially at the chain’s outer ends deserves no ink. **open** turns the row into the matrix $M_1 M_2 M_3$, and **periodic** (sugar for **west=trace**, **east=trace**) closes the row on itself. Same atoms, same bonds; only the boundary signature moves.

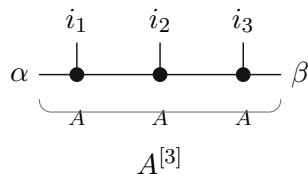
The default is a statement, not an omission. The deleted 0.7 dialect drew west and east stubs unbidden and sealed them with **none**; the kernel mints a leg only on declaration, so **none** now changes nothing and merely says so. That reading died with the old front end — if you sealed stubs once, stop: there are no stubs to seal.

EXERCISE 3.4 Draw the coefficient of the periodic matrix product state — the scalar $\text{tr}(M_1 M_2 M_3)$ — and say what its boundary signature is.

3.4 A mark over a run

Blocking is the coarse-graining step of every renormalization argument in the matrix-product literature: three sites read as one blocked tensor $A^{[3]}$. The contraction does not change; a mark speaks over it.

EXAMPLE 3.4 · blocking three sites⁴



$$(M_1 M_2 M_3)_{\alpha\beta}$$

³ Left to right: the outer bonds close trivially and nothing is exposed; both ends open and the row is a matrix; the trace returns the row to itself and the picture is a scalar. The quoted signature is the open panel’s; the other two are empty.

`kernel-boundary|signature=open:e,
open:w`



$$A^{[3]} = A^{i_1} A^{i_2} A^{i_3}$$

⁴ The bracket speaks from the south, past the site names and their ink. The signature is Example 3.1’s, unchanged: a mark owns no topology.

`kernel-boundary|signature=open:e,
open:w, phys:n, phys:n, phys:n`

```
% Formula: the blocked word A^{[3]}; the mark adds no index.
\begin{tenkz}[cols=3]
  \tn[ports={180:virtual:$\alpha$, 0:virtual, 90:physical:$i_1$}]{A} &
  \tn[ports={180:virtual, 0:virtual, 90:physical:$i_2$}]{A} &
  \tn[ports={180:virtual, 0:virtual, 90:physical:$i_3$}]{A}
  \tnmark[form=bracket]{(1,1) .. (1,3)}{A^{[3]}}
\end{tenkz}
```

`\tnmark` annotates a target without touching the contraction. The target here is a selector: $(1,1)$ names a cell of the frame, and the two-dot range $(1,1) \dots (1,3)$ names the records between two cells in the frame's own order. A bracket that names no side speaks from the south, where a brace under a span belongs. Compare the margin with Example 3.1: the signature has not moved, because marks never own topology — delete every mark and no contraction changes.

The range is written with two dots, never a hyphen, because generated record names carry hyphens of their own: a cluster member is named `a-2-2`, and a hyphen range could not be told from it.



EXERCISE 3.5 Restate the same blocked run as a tinted enclosure whose label sits below the contour.

3.5 An equation under audit

The fifth idea is the one the others were preparing: whole pictures compose into an equation, and the equation is an audit scope. The identity is the gauge freedom of the matrix product state — two tensors, $B^i = X A^i X^{-1}$, generating the same state.

EXAMPLE 3.5 · a gauge identity, audited⁵

$$B^i = X A^i X^{-1}$$

⁵ The rings are matrices on the wire and add no indices. Both panels expose one west stub, one east stub, and one physical leg, so the audit records the relation as equal.

```
check|scope=1|relation=1|
result=equal|signature=open:e,
open:w, phys:n
```

```
% Formula: B^i = X A^i X^{-1}; one relation, audited.
\[
\begin{tenkz}[check=signature]
  \begin{tenkz}[cols=1]
    \tn[ports={180:virtual, 0:virtual, 90:physical:$i$}]{B}
  \end{tenkz}
  =
  \begin{tenkz}[cols=3]
    \tn[skin=ring, ports={180:virtual, 0:virtual}]{X} &
    \tn[ports={180:virtual, 0:virtual, 90:physical:$i$}]{A} &
    \tn[skin=ring, ports={180:virtual, 0:virtual}]{X^{-1}}
  \end{tenkz}
\end{tenkz}
```

`tenkz` places its panels around the mathematics between them — the equals sign is ordinary TeX — under one shared metric, so a glyph is one size on both sides of an equality. Then it audits: every panel carries a boundary signature, and a relation requires

its two sides to expose the same one. `check=signature` names that audit (it is also the default: writing it states what is already running), and the margin quotes the `check` event the equation wrote beside each panel's boundary line. A diagrammatic identity in this language is not a juxtaposition that looks right; it is a comparison that ran.

What does a failed comparison look like? Drop the physical leg of B and the compiler answers:

A DELIBERATE REFUSAL 3.1 · the audit catches a dropped leg⁶

```
% Refused: B lost its physical leg, so the relation's two
% sides no longer expose the same boundary.
\[
\begin{tenkzeq}[check=signature]
\begin{tenkz}[cols=1]
\tn[ports={180:virtual, 0:virtual}]{B}
\end{tenkz}
=
\begin{tenkz}[cols=3]
\tn[skin=ring, ports={180:virtual, 0:virtual}]{X} &
\tn[ports={180:virtual, 0:virtual, 90:physical:$i$}]{A} &
\tn[skin=ring, ports={180:virtual, 0:virtual}]{X^{-1}}
\end{tenkz}
\end{tenkzeq}
\]
```

[TKZ-EQ-SIGNATURE] sides 1 and 2 expose unequal boundaries: (open:e, open:w) versus (open:e, open:w, phys:n).

Read refusals as language, not as accidents: the code names the check that failed, and the message prints both signatures so the missing entry is visible at once. The correction is the port list of Example 3.5's B .

Relations are one of three joiner classes. A sum also requires equal signatures; a product instead contracts the facing cut, east entries of the left factor cancelling west entries of the right, and exposes the rest. The comparison can be taken up to bundling — `check={signature, modulo=bundles}` reads a bundled index and its members as one entry — and a relation whose two sides are known, unequal, and equal anyway for a reason the diagram does not draw carries a recorded opt-out, `off={<relation>: <reason>}`, written to the event stream beside the waiver. The audit cannot be switched off silently.



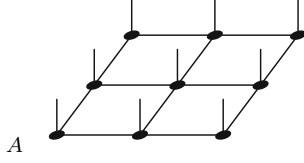
EXERCISE 3.6 Why does `check={signature, modulo=member}` refuse?

3.6 A sheet of tensors

The chain taught the whole alphabet; the second dimension only adds a frame. The object is the projected entangled pair state (PEPS), the wavefunction of a two-dimensional system: one five-index tensor A per lattice site, four virtual indices of bond dimension D contracted with the four neighbours, one physical index of dimension d standing free above the sheet, and the tensor A generating the state $|\psi(A)\rangle$.

⁶ The left panel now exposes two indices, the right three. The correction is one token: restore the port 90:physical: i on B .

EXAMPLE 3.6 · a PEPS on a 3×3 patch⁷



```
% Formula: the PEPS  $|\psi(A)\rangle$  generated by one tensor A.
% Ink: in-sheet bonds contract; each transverse leg is one
% physical index.
\begin{tenkz}[lattice={3x3}, frame=plane, physical=up]
  \tn{} & \tn{} & \tn{} \\
  \tn{} & \tn{} & \tn{} \\
  \tn{} & \tn{} & \tn{}
  \tn[label pos=225]{A} & \tn{} & \tn{}
\end{tenkz}
```

Two spellings put the sheet on the page. `lattice={3x3}` is sugar for three wire rows of three columns, and the cells contract with their neighbours exactly as chain cells do — those are the virtual bonds of the sheet. `frame=plane` projects the grid, and `physical=up` grows one physical leg per site along the plane’s transverse axis — the direction that stands vertically on the page and belongs to the frame itself. No side policy is stated, so no virtual index is exposed: the patch has open boundary conditions, and its boundary signature is the nine physical legs of the wavefunction.

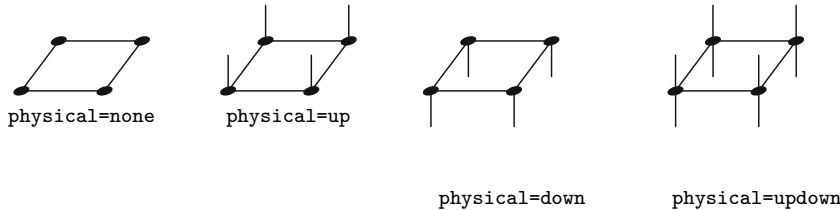
A plane owns three directions, and only two lie in the sheet. `ports={90:physical}` on a plane atom is the in-sheet north face, not the transverse leg; the physical axis is reached only through the picture policy. A PEPS tensor is four numeric ports plus the policy — never a fifth angle compensating for the projection.

EXERCISE 3.7 Draw one PEPS tensor by itself — the five-index A^i_{uldr} : four open virtual ports labelled u, l, d, r in the sheet, and the physical leg above.

3.7 One sheet, four readings

The same sheet is four objects of the two-dimensional literature, and one enum word turns between them: a fully contracted network is a partition function Z ; legs up is the state $|\psi\rangle$; legs down is its conjugate $\langle\psi|$; legs both ways is a projected entangled pair operator O , the PEPO.

EXAMPLE 3.7 · one sheet, four readings⁸



$|\psi(A)\rangle$

⁷ Every dot is the same tensor A . The in-sheet bonds are the contracted virtual indices; each transverse leg is one physical index, and the signature lists the nine of them.

```
kernel-boundary|signature=phys:n,
phys:n, phys:n, phys:n, phys:n,
phys:n, phys:n, phys:n, phys:n
```



$Z, |\psi\rangle, \langle\psi|, O$

⁸ Left to right: no physical legs, a fully contracted scalar; legs up, the state; legs down, its conjugate; legs both ways, the PEPS operator. The quoted signature is the legs-up panel’s.

```
kernel-boundary|signature=phys:n,
phys:n, phys:n, phys:n
```

```
% Formula: Z, |psi>, <psi|, 0 -- one sheet, four leg policies.
\begin{tenkz}[lattice={2x2}, frame=plane, variant]
  \tn{} & \tn{} \\\
  \tn{} & \tn{}
\end{tenkz}
```

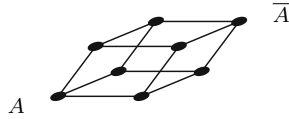
One family, one figure again: the four panels share every token except the word under them, and the family is enumerated in full. The signature tells a ket from a bra before any render is looked at: a leg above the sheet writes `phys:n`, a leg below writes `phys:s`, so the record stream distinguishes the state from its dual, and the PEPO panel exposes both lists at once.

EXERCISE 3.8 A PEPS with periodic boundary conditions lives on a torus. The one word `surface=torus` closes all four sides of the patch — sugar for `trace` on each of `west=`, `east=`, `north=`, and `south=`. What does it change in the boundary signature of Example 3.6, and what in the network?

3.8 The double layer

Every expectation value in the PEPS literature is computed in the double layer: the conjugate sheet laid over the state, each site's physical index contracted with its conjugate copy. The norm $\langle\psi|\psi\rangle$ is the double layer with nothing inserted.

EXAMPLE 3.8 · the norm as a double layer⁹



```
% Formula: <psi|psi> -- the bra sheet contracted onto the
% ket sheet through every physical index.
\begin{tenkz}[lattice={2x2}, planes, bonds=none]
  \tn[at=(2,1,1), label pos=225]{A}
  \tn[at=(1,2,2), label pos=45]{\overline{A}}
  % ket-sheet virtual bonds
  \tnwire{(1,1,1)}{(1,2,1)} \tnwire{(2,1,1)}{(2,2,1)}
  \tnwire{(1,1,1)}{(2,1,1)} \tnwire{(1,2,1)}{(2,2,1)}
  % bra-sheet virtual bonds
  \tnwire{(1,1,2)}{(1,2,2)} \tnwire{(2,1,2)}{(2,2,2)}
  \tnwire{(1,1,2)}{(2,1,2)} \tnwire{(1,2,2)}{(2,2,2)}
  % ket-bra physical pairings
  \tnwire{(1,1,1)}{(1,1,2)} \tnwire{(1,2,1)}{(1,2,2)}
  \tnwire{(2,1,1)}{(2,1,2)} \tnwire{(2,2,1)}{(2,2,2)}
\end{tenkz}
```

`planes` is sugar for one plane frame with a two-member basis — a ket sheet and a bra sheet — so both copies live in one frame and every site has a member address: $(r,c,1)$ is the ket copy of cell (r,c) , $(r,c,2)$ the bra copy. A multi-member basis draws no bond it is not told — `bonds=none` says so — and the body declares each one: eight virtual bonds, sheet by sheet, then four ket–bra pairings. A pairing carries the type neither of its ends can state, because a member address names a lattice place and a place is not

$\langle\psi|\psi\rangle$

⁹ Two 2×2 sheets in one frame: A marks the ket sheet, $\overline{A}$ the bra copy above it. Each short diagonal is one ket–bra physical contraction; nothing stays open, and the empty signature says scalar.

kernel-boundary|signature=

a port; the frame knows its transverse axis, types the wire physical, and the record stream says so.

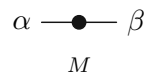
The two sheets interleave in projection, and several bonds pass close by sites of the other sheet. Nothing contracts there: coordinate coincidence neither identifies records nor creates an edge, and the only contractions are the twelve wires the body declares.

EXERCISE 3.9 Insert a one-site operator. The expectation value $\langle \psi | O | \psi \rangle$ stands the ring O on one ket–bra pairing: name the pairing wire and put the operator halfway along it.



3.9 Answers to the exercises

ANSWER 3.1 ¹⁰



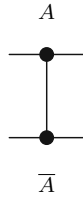
```
% Formula: the matrix M with both virtual indices open.
\begin{tenkz}[cols=1]
  \tn[ports={180:virtual:$\alpha$, 0:virtual:$\beta$}]{M}
\end{tenkz}
```

$M_{\alpha\beta}$

¹⁰ Two typed ports with no partners: the matrix keeps two open virtual indices and nothing else.

kernel-boundary|signature=open:e,
open:w

ANSWER 3.2 ¹¹



```
% Formula: E = sum_i A^i tensor conj(A^i).
\begin{tenkz}[rows={ket,bra}, cols=1, west=open, east=open]
  \tn[label pos=90]{A} \ \
  \tn[label pos=270]{\overline{A}}
\end{tenkz}
```

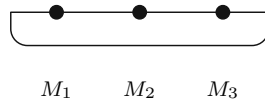
$$E = \sum_i A^i \otimes \overline{A^i}$$

¹¹ The frame contracts the ket-bra physical pair; all four virtual ends stay open, so E is a map on the doubled space.

kernel-boundary|signature=open:e,
open:e, open:w, open:w

ANSWER 3.3 A closed wire is a smooth cycle, so it has no ends to give; every other wire has exactly two. The spelling states both at once and is refused with the coded error [TKZ-LANG-WIRE-ARITY]: *a closed wire takes no positional ends; every other wire takes exactly two*. Keep `closed` for a free loop, or keep the two ends for a bond.

ANSWER 3.4 ¹²

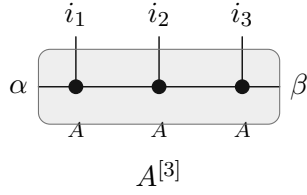


```
% Formula: tr(M_1 M_2 M_3); sugar: boundary=periodic
% expands to west=trace, east=trace.
\begin{tenkz}[rows={wire}, cols=3, boundary=periodic]
  \tn{\$M_1\$} & \tn{\$M_2\$} & \tn{\$M_3\$}
\end{tenkz}
```

$\text{tr}(M_1 M_2 M_3)$

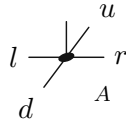
¹² The trace returns the row to itself; no index is left open, and the empty signature says scalar.

kernel-boundary|signature=

ANSWER 3.5 ¹³

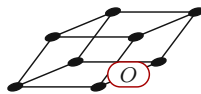
```
% Formula: the blocked word inside one tinted contour.
\begin{tenkz}[cols=3]
  \tn[ports={180:virtual:$\alpha$, 0:virtual, 90:physical:$i_1$}]{A} &
  \tn[ports={180:virtual, 0:virtual, 90:physical:$i_2$}]{A} &
  \tn[ports={180:virtual, 0:virtual:$\beta$, 90:physical:$i_3$}]{A}
  \tnmark[form=enclosure, tint,
    label pos=270]{(1,1) .. (1,3)}{$A^{[3]}$}
\end{tenkz}
```

ANSWER 3.6 The audit refuses words it does not know rather than passing over them, because a spelling the audit ignores is a comparison you believe is running and is not. The coded error is [TKZ-EQ-CHECK-WORD]: *the audit does not know ‘modulo=member’; check= states the audit (signature), the equivalence it is taken up to (modulo=bundles), and its recorded opt-outs (off={<relation>: <reason>})*. The one admitted equivalence word is **bundles**.

ANSWER 3.7 ¹⁴

```
% Formula: the five-index PEPS tensor A^{i_{uldr}}.
\begin{tenkz}[rows={wire}, cols=1, frame=plane,
  bonds=none, physical=up]
  \tn[ports={0:virtual:$r$, 90:virtual:$u$,
    180:virtual:$l$, 270:virtual:$d$},
    label pos=315]{A}
\end{tenkz}
```

ANSWER 3.8 Nothing moves in the signature, and six wires appear in the network. Each **trace** closes virtual bonds the patch never exposed, so the boundary signature still reads nine physical legs; the network gains one wrap wire per row and per column — **wrap-1** through **wrap-3** and **wrap-col-1** through **wrap-col-3** — closing the sheet into a torus. The signature is the boundary, not the topology: the open patch and the periodic state expose one list and are different networks.

ANSWER 3.9 ¹⁵ $A^{[3]}$

¹³ The enclosure strokes the offset hull of the selection and tint lays ink over the paper inside it; the physical legs pierce the contour, and the signature still does not move.

```
kernel-boundary|signature=open:e,
open:w, phys:n, phys:n, phys:n
```

 $A^{i_{uldr}}$

¹⁴ Four open in-sheet virtual indices and one transverse physical leg: the five-index PEPS tensor. The two receding faces expose their in-sheet bearings in the signature.

```
kernel-boundary|
signature=open:233.130102,
open:53.130102, open:e, open:w,
phys:n
```

 $\langle\psi|O|\psi\rangle$

¹⁵ The ring stands halfway along the named ket–bra pairing: a one-site operator between the state and its conjugate. The signature stays empty — the expectation value is a scalar.

```
kernel-boundary|signature=
```

```
% Formula: <psi|0|psi> -- 0 on one site's physical index.
\begin{tenkz}[lattice={2x2}, planes, bonds=none]
  \tnwire{(1,1,1)}{(1,2,1)} \tnwire{(2,1,1)}{(2,2,1)}
  \tnwire{(1,1,1)}{(2,1,1)} \tnwire{(1,2,1)}{(2,2,1)}
  \tnwire{(1,1,2)}{(1,2,2)} \tnwire{(2,1,2)}{(2,2,2)}
  \tnwire{(1,1,2)}{(2,1,2)} \tnwire{(1,2,2)}{(2,2,2)}
  \tnwire{(1,1,1)}{(1,1,2)} \tnwire{(1,2,1)}{(1,2,2)}
  \tnwire{(2,1,1)}{(2,1,2)}
  \tnwire[name=po]{(2,2,1)}{(2,2,2)}
  \tn[skin=ring, at=on po 0.5]{0}
\end{tenkz}
```

4 Recipes

Each recipe changes one semantic decision. Start from the nearest tutorial example and add only the policy the mathematics requires.

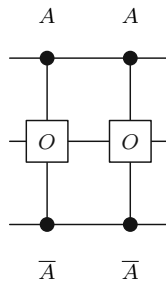
4.1 Keep a map open; close a scalar

A side exposes indices only when it says so. A picture that states no side policy draws no side leg and carries the empty boundary signature on those sides, so a map keeps its ends by writing **boundary=open** – or **west=open** for one side alone – and **boundary=none** is the explicit spelling of what the picture already does. The frame's grid places atoms; it does not claim that every rim cell owns a dangling index, which is why a circuit pyramid or a single four-legged junction collects no legs it did not ask for.

The rest of the alphabet: use **boundary=periodic** for a same-row trace. Use **west=cup** or **east=cup** to connect adjacent layers. These closures can have the same count of open ends and still mean different operations.

4.2 Stack ket, operator, and bra

EXAMPLE 4.1 · an expectation-value window¹



$$\sum_{s,s'} A^s \otimes O^{ss'} \otimes \bar{A}^{s'}$$

¹ The rows declare their roles.

Vertical connections contract physical indices; six horizontal stubs keep the window a map.

kernel-boundary|signature=open:e,
open:e, open:e, open:w, open:w,
open:w

```
% Formula: sum_{s,s'} A^s tensor O^{ss'} tensor conjugate(A^{s'}).
% Labels: the ket names speak north, away from the physical bonds.
\begin{tenkz}[rows={ket,op,bra}, cols=2, west=open, east=open]
  \tn[label pos=90]{A} & \tn[label pos=90]{A} \\
  \tn[skin=mpo]{O} & \tn[skin=mpo]{O} \\
  \tn{\overline{A}} & \tn{\overline{A}}
\end{tenkz}
```

4.3 Block a run without changing topology

A bracket mark over a cell range (`\tnmark` with `form=bracket`) declares a brace or measured group over existing atoms; `skin=dots` declares an ellipsis cell whose wires pass through. Neither creates a contraction. A wide atom is appropriate only when the formula has already replaced the run by one blocked tensor.

4.4 Distinguish a role from a species

Use `role=` for a fixed semantic role such as `operator` or `marked`. Declare `species=` at setup scope when a paper has a finite family whose members keep one hue across figures. Labels distinguish objects of the same kind; color does not encode identity. A role may change style, never topology.

4.5 Stand a label on an open leg

An open leg is a wire like any other and it carries a name. A side policy names its legs after the side and the row or column they leave – `open-west-1`, `open-south-col-3` – the `physical=` policy names each of its legs after the face and the cell it grows from, `leg-n-1-1` and `leg-s-2-3`, and an authored open end takes the name its `\tnwire` declares. Address a fraction of that name to put a label or a bead on the leg:

```
\begin{tenkz}[rows={wire}, cols=2, west=open, east=open]
  \tn{A} & \tn{B}
  \tnmark[form=label, label pos=90]{on open-west-1 0.5}{\alpha}
\end{tenkz}
```

The station is read along the leg the picture draws. A leg has one end on a glyph and one free tip, and fraction zero is the end the wire is written from, so a leg written from a port to `open e` runs outward while one written from `open w` to a port runs inward. A policy leg runs outward from the glyph, so `on leg-n-1-1 0` is at the north face of the site in row one, column one, and `on leg-n-1-1 1` is at its free tip. Stating a row's legs once with `physical=` therefore costs nothing: the same name serves the crossing grammar and the address.

4.6 Say which bond is the one under discussion

A lattice draws its bonds from the row and column policy, and every one of them is the same black contraction. A figure that distinguishes one edge – the edge being blocked, the bond being cut – restates that bond in the body. An index wire between two adjacent cells is that pair's bond: the frame mints no second wire under it, and the wire's species, stroke, and route are the bond's.

```
\tndeclare{species}{blocked-edge}{hue=source:red!65!black}
\begin{tenkz}[rows={wire,wire}, cols=3]
  \tn{} & \tn{} & \tn{}\\
```

```

\tn{} & \tn{} & \tn{}
\tnwire[species=blocked-edge]{(1,1)}{(1,2)}
\end{tenkz}

```

The restatement is the same contraction, so the count of internal bonds and the exposed boundary do not move; only the ink does. Either spelling of the pair reaches it – the two cells, as above, or the two typed ports `a.0` and `b.180` on named atoms. `bonds=none` plus a hand-written wire for every pair states the same picture and is not a spelling anyone should use. A travelling string is not a restatement: it carries an operator index, so it is drawn over the lattice and retires nothing.

4.7 Send a physical index sideways

An index type belongs to the port that carries it. `ports=` types each face on its own, so `0:physical` sends a physical index east exactly as `0:virtual` sends a virtual one, and a map drawn as a single atom types its two faces differently:

```

\begin{tenkz}[rows={wire}, cols=1, bonds=none]
\tn[at=(1,1), skin=box, ports={180:virtual:V, 0:physical:W}]{A}
\end{tenkz}

```

`physical=` is a picture policy, not the only source of a physical index: it gives every frame-cell atom one outward physical port on the frame's own outward face. The two spellings stand side by side, and neither claims the other's face.

The policy earns its place on a uniform row: state it once and every site takes the leg, with the rare exception refusing it by `physical=none`. It costs on a sparse row, where a single site among many carries the index – every other site must then refuse it, and the refusals outrun the port list they replaced. Reach for the policy when two or more sites share the leg; write the ports when only one does:

```

\begin{tenkz}[rows={wire}, cols=5]
\tn{} & \tn{} & \tn[ports={180:virtual, 0:virtual, 90:physical:$i$}]{ } &
\tn{} & \tn{}
\end{tenkz}

```

4.8 Keep a physical index out of a projected sheet

There are five directions at a PEPS tensor, but only four lie in the sheet. The four numeric ports below follow the projected axes. The picture policy supplies the remaining, page-vertical direction.

```

\begin{tenkz}[rows={wire,wire,wire}, cols=3,
              frame=plane, bonds=none, physical=up]
\tn[at=(2,2),
     ports={0:virtual,90:virtual,180:virtual,270:virtual}]{A}
\end{tenkz}

```

Do not add a fifth numeric port to straighten the physical leg. A numeric face always belongs to the sheet; `physical=up` belongs to the plane’s independent transverse axis.

Physical indices belong to sites named by frame addresses. A mark placed at a midpoint, on a wire, or at a crossing is a point constructed from the existing geometry, not another site; the picture policy therefore gives it no physical index. If such a mark represents an additional tensor with a physical index, declare that port explicitly with `ports=`.

4.9 Draw fusion trees; leave commutative diagrams to `tikz-cd`

Use `\tntree` for a parenthesized fusion word. A commutative diagram composes morphisms rather than contracting tensor indices; it belongs to `tikz-cd`, outside this language. A `\tntree` atom is a self-contained box and sits directly in a `tikz-cd` cell.

4.10 Put a picture in a sentence

A picture enters surrounding TeX as a displayed term today. The inline composition wrapper – one picture behaving as a running-math atom, with the metric sensing its mathematical context – is contract work that has not landed; when it does, it will be a composition spelling, not a key: no record inside the picture acquires an “inline” property.

5 Canonical grammar reference

The executable language registry generates this chapter. It inventories environments, commands, keys, scopes, value types, defaults, diagnostics, and standalone examples. The registry also drives parser registration, linting, and the public-symbol census.

Command	Class	Why this is a command
<code>\tn</code>	atoms	An atom changes the object record, so it is a command rather than a key.
<code>\tn [object keys]{label}</code>		
Example: <code>tex/tenkz/examples/api/tn.tex</code> . Representative failure: An unknown object key is a hard error.		
<code>\tnfuse</code>	atoms	A fusion atom changes wire arity.
<code>\tnfuse [object keys]{label}</code>		
Example: <code>tex/tenkz/examples/api/tnfuse.tex</code> . Representative failure: A nonpositive span is a hard error.		
<code>\tntree</code>	atoms	A fusion tree is one structured atom.
<code>\tntree [keys]{word}</code>		
Example: <code>tex/tenkz/examples/api/tntree.tex</code> . Representative failure: An unbalanced fusion word is a hard error.		
<code>\tnset</code>	setup-extension	Setup changes document or group policy outside any one picture.
<code>\tnset {setup keys}</code>		

Example: `tex/tenkz/examples/api/tnset.tex`. Representative failure: An unknown setup key or enum value is a hard error.

`\tndeclareatom` setup-extension A declaration adds a typed atom spelling to the grammar.
`\tndeclareatom {\command}{skin=value, ports={face:type}}`

Example: `tex/tenkz/examples/api/tndeclareatom.tex`. Representative failure: An unknown face, type, skin, or duplicate command is a hard error.

`\tnwire` connections- A wire is one typed index line;
 closures bonds and strings are its kinds.
`\tnwire [wire keys]{end}{end}`

Example: `tex/tenkz/examples/api/tnwire.tex`. Representative failure: A closed wire with positional ends is a hard error.

`\tnmark` regions- A mark annotates without
 annotations owning topology.
`\tnmark [mark keys]{target}{label}`

Example: `tex/tenkz/examples/api/tnmark.tex`. Representative failure: An unknown mark form is a hard error.

`\tngroup` composition A group transforms a sub-
 diagram as one object.
`\tngroup [picture keys]{body}`

Example: `tex/tenkz/examples/api/tngroup.tex`. Representative failure: An unknown picture key is a hard error.

`\tndeclare` setup-extension The one extension door: atom,
 skin, or species.
`\tndeclare {class}{name}{keys}`

Example: `tex/tenkz/examples/api/tndeclare.tex`. Representative failure: An unknown declaration class is a hard error.

`\tnbond` connections- Sugar: a bond is a wire of kind
 closures index.
`\tnbond [wire keys]{end}{end}`

Example: `tex/tenkz/examples/api/tnbond.tex`. Representative failure: An unknown wire key is a hard error.

`\tenkzkernel` setup-extension Rebinds the kernel surface;
 tenkz.sty calls it at load, so a
 document's own call is inert.
 Sunset 1.0.

`\tenkzkernel`

Example: `tex/tenkz/examples/api/tenkzkernel.tex`. Representative failure: The package binds the kernel surface at load; a document's own call rebinds the same meanings.

`\tnprose` regions- Sugar: recorded prose in place of
 annotations a diagram.
`\tnprose {text}`

Example: `tex/tenkz/examples/api/tnprose.tex`. Representative failure: Prose on a checked relation without an opt-out is a hard error.

Scope	Key	Type	Default	Meaning
object	<code>role</code>	semantic-role	empty	Semantic theme role.
object	<code>species</code>	declared-name	empty	Semantic species.
object	<code>tree style</code>	enum(wire ribbon)	wire	Fusion-tree strand style.
atom-declaration	<code>skin</code>	silhouette-name	dot	Declared atom silhouette.
atom-declaration	<code>ports</code>	typed-port-list	empty	Declared atom ports.

kernel-picture	rows	row-list	wire	Chain row topology; non-wire rows populate.
kernel-picture	cols	positive-integer	3	Cells per row.
kernel-picture	frame	frame-spec	flat	Closed flat, plane, or circle frame word.
kernel-frame	basis	basis-spec	origin singleton	Ordered row-kind members at east/north quarter-pitch offsets.
kernel-picture	west	side-policy	none	West side policy.
kernel-picture	east	side-policy	none	East side policy.
kernel-picture	north	side-policy	none	North side policy.
kernel-picture	south	side-policy	none	South side policy.
kernel-picture	trace	trace-spec	empty	Cells or physical ports closed to themselves.
kernel-picture	open	cell-set	empty	Cells opened by declaration.
kernel-picture	bonds	bond-policy	grid	Frame-generated adjacent contraction.
kernel-picture	align	row	midline	The wire axis panels align on.
kernel-picture	size	size-class	from math style	Size class of the metric context.
kernel-picture	metrics	enum(compact)	empty	Named metric profile of the picture's pitch.
kernel-picture	check	check-spec	signature	The tenkz audit: selection, equivalence, recorded opt-outs.
kernel-picture	sandwich	flag	false	Ket-over-bra row preset.
kernel-picture	boundary	enum(open none periodic)	none	All-side boundary word.
kernel-picture	lattice	pair	empty	RxC wire-row lattice preset.
kernel-picture	ring	integer	empty	Circle-frame ring preset.
kernel-picture	surface	identifier	empty	Torus closure preset.
kernel-picture	planes	flag	false	Plane frame with ket and bra basis members.

kernel-picture	physical	physical-policy	none	Expander: frame-owned outward physical port per frame-cell atom; geometric overlays are excluded.
kernel-atom	skin	silhouette- name	theme default	Declared silhouette.
kernel-atom	wide	positive-integer	1	Cells spanned.
kernel-atom	wires	positive-integer	1	Wire rows spanned.
kernel-atom	at	address	next chain cell	Placement address.
kernel-atom	name	identifier	generated	Addressable name.
kernel-atom	ports	typed-port-list	from skin	Typed faces and slots.
kernel-atom	species	declared-name	empty	Semantic species.
kernel-atom	pairing cross	indexed- crossing-list	empty	Per-instance crossings of generated skin pairings.
kernel-atom	size	size-class	m	Per-atom metric size class.
kernel-atom	label pos	angle	auto	Label quadrant.
kernel-atom	nudge	pair	zero	Quarter-pitch displacement.
kernel-atom	void	void-policy	unset	Hole or removed site.
kernel-atom	physical	physical-policy	unset	Site's own answer to the picture's physical policy.
kernel-atom	conjugate	flag	false	Tombstoned; stored, read by nothing.
kernel-atom	cluster	pair	empty	Sunset 1.0. Expander: RxC addressable sub-atom group.
kernel-atom	role	semantic-role	empty	Prelude species spelling.
kernel-wire	kind	enum(index string pairing)	index	Bond, travelling string, or generated skin pairing.
kernel-wire	route	enum(straight orth arc)	straight	Path family.
kernel-wire	via	address-list	empty	Waypoints.
kernel-wire	bend	number	metric	Arc bend factor.
kernel-wire	weight	enum(single double bundle=n)	single	Strand multiplicity.
kernel-wire	species	declared-name	empty	Semantic species.
kernel-wire	wind	pair	zero	Homotopy class on traced frames.

kernel-wire	around	address-list	empty	Silhouette detour targets.
kernel-wire	cross	crossing-list	empty	Declared over and under crossings.
kernel-wire	crossing	enum(over under alternate alternate=over alternate=under)	empty	Default crossing order at every crossing the string makes.
kernel-wire	dir	direction- policy	none	Direction mark of a directed index.
kernel-wire	stroke	enum(solid dashed dotted)	solid	Declared stroke pattern of the rail.
kernel-wire	name	identifier	generated	Addressable name.
kernel-wire	closed	flag	false	A smooth closed cycle; takes no ends.
kernel-mark	form	enum(bracket brace-above brace-below enclosure cut band label prose)	label	Mark form.
kernel-mark	slot	semantic-slot	selected	Semantic tint.
kernel-mark	species	declared-name	empty	Semantic species.
kernel-mark	inset	number	0	Nested- contour step.
kernel-mark	label pos	angle	auto	Label quadrant.
kernel-mark	nudge	pair	zero	Quarter-pitch displacement.
kernel-mark	name	identifier	generated	Addressable name.
kernel-mark	tint	flag	false	Interior tint under the contour.
kernel-setup	pitch	length	em-relative	Base metric.
kernel-setup	sizes	size-table	bundled table	Size-class table.
kernel-setup	strict	flag	false	Rejects sugar; benchmark and CI set it.
kernel-setup	theme	identifier	house	Semantic ink theme.
kernel-declare	hue	hue-source	house cycle	Species ink; source:* reproduces a cited palette.
kernel-declare	base	identifier	empty	Skin the declaration extends.
kernel-declare	pairings	port-pair-list	empty	Skin-internal wires between own ports.

Prelude class	Name	Declaration
---------------	------	-------------

skin	mpo	base=box
skin	pill	base=roundrect

Canonical sources use only the spellings in the main generated tables. Compatibility aliases are isolated in Appendix A; they never appear in tutorials or benchmark cases.

6 Atom extension and house style

The stock atom skins are shared across layouts. A dot is an unnamed tensor site, a box or pill carries an inscribed label, an MPO has physical input and output faces, and a ring is a matrix on an existing wire. The semantic kind, typed ports, and visual skin are separate data.

6.1 Declare one typed atom

Use `\tndeclare` only when the stock atom vocabulary cannot express a recurring mathematical kind:

```
\tndeclare{atom}{\langle command \rangle}
{skin=\langle dot/box/pill/mpo \rangle,
 ports={\langle face:type \rangle,...}}
```

```
\tndeclare{atom}{\tnprojector}{
  skin=box,
  ports={180:virtual, 0:virtual, 90:physical}
}
\begin{tenkz}[cols=1] \tnprojector{P} \end{tenkz}
```

A face is an angle in the atom's own axes: 180 and 0 are the two ends of the wire, 90 is above and 270 below. Types are **virtual** and **physical**. The declaration creates a one-label atom command with optional per-use cell keys. Its port list is complete: an anchor without a type is not an extension contract.

¹

There is no `\tndefine`. Naming a repeated whole figure would hide the grammar benchmark and create a second topology source. Compose ordinary TeX macros only outside canonical benchmark bodies, and keep their expansion in public *tenkz* syntax.

¹ *Why a command, not a key?* The declaration adds a new atom kind and therefore a new word in the atom class. A skin or role change on an existing atom is a key instead.

6.2 Extension gate

A new key needs at least three manifested RMP consumers. A new command needs a new grammatical class. Every accepted capability also needs a registry entry, a standalone example of roughly fifteen lines or fewer, a stable coded diagnostic, a negative test, and a teachability review.

6.3 House style

Use `\tnset` for document-wide metric or semantic policy and picture keys for local policy. Themes may rebind semantic hues and typography; they do not add topology. Use roles for meaning and labels for identity. Never assign arbitrary colors to distinguish tensors of the same kind.

Labels occupy reserved bands outside measured silhouettes. If an entire picture needs more room, use the documented metric or layer separation. A manual coordinate nudge is the last resort and is never canonical benchmark syntax. The geometry stage measures occupied ink and separates it with one daylight constant.

7 The RMP benchmark field guide

The manual teaches the language. The separate *tenkz RMP benchmark* stress-tests it against 130 agreed targets: 92 published targets in paper order and 38 author-workbench targets in source order. It is built from `docs/tenkz/rmp-benchmark.tex`; this manual does not duplicate its target catalogue.

Each target lives in one preamble-free source file under `tests/tenkz/rmp/`. That exact file is wrapped for standalone regression, input by the book, rendered by the web/SVG pipeline, and printed verbatim on the target's spread. No source body is copied into a driver, manifest, generated wrapper, or manual.

Every target header states its formula, ink, boundary, source anchors, and capabilities. The verso page prints that contract and complete line-numbered source. The recto page gives the vector rendering, audited boundary signature, dimensions, digests, and print/web verdict. Digest changes make a prior verdict stale automatically.

Use the shared frontend:

```
python3 scripts/tenkz_rmp.py check --id <target>
python3 scripts/tenkz_rmp.py check --section <section>
python3 scripts/tenkz_rmp.py check --all
python3 scripts/tenkz_rmp.py book --all
python3 scripts/tenkz_rmp.py render --all
python3 scripts/tenkz_rmp.py compare --all --source-root /absolute/path/to/author-source-tree
```

The benchmark style controls typography only. It does not extend *tenkz*. The separate author-source tree is not part of this repository. Reference comparisons are compiled temporarily from its mapped source blocks; prebuilt PDFs are not design evidence. Targets without author drawing source do not pretend that a reference implementation exists. They carry the status `paper-context-only`.

8 Troubleshooting and maintenance

8.1 Read meaning before ink

Open the job's `.tnlog` and find the picture's coded diagnostic and boundary event. A matrix needs two open virtual indices, a doubled-space map needs four, and a scalar needs none. Compare an equation side by side, not merely by total count.

Symptom	First correction
A map rendered as a scalar	State boundary=open , or type the open ports; an unstated side draws no leg.
A trace looks like a layer cup	Use boundary=periodic for same-row closure; use side cup only between adjacent layers.
A label touches ink	Restore a documented metric or label band; do not hide the problem with an undocumented coordinate offset.
Requested ink disappears	Treat it as an implementation defect. Unsupported ink must be a hard coded error, never a warning followed by omission.

Brace any key value containing a comma or a second equals sign. A conjugate is label mathematics and a wire direction, not an atom flag. A genuine hole is **void=open**, because its incident indices survive; **void=sealed** removes the site and its bonds.

8.2 Maintainer pipeline

The internal flow is

```
language -> model -> style/atoms -> geometry -> kernel
          layout -> rendering -> events.
```

Language owns declarations and diagnostics. Model owns normalized records. Style owns metrics, atom descriptors, label policy, ports, and silhouettes. Geometry owns frames, measurement, daylight, routing, and crossings. Dialect layout owns only genre-specific placement. Rendering emits ink without parsing syntax or changing topology. Events serialize the resolved model used by rendering.

`tenkz.sty` is a logic-free load map. Each internal file begins with a five-line contract naming its input, output, owned state, invariants, and next stage. Private control sequences follow one owner-prefixed convention; the symbol census rejects accidental exports.

Build and debugging commands live in `HACKING.md`; the full ownership contract lives in `ARCHITECTURE.md`. After every meaningful change, compile, lint, audit the versioned event stream, render, and inspect the affected pages. Exit codes do not inspect diagrams.

A Compatibility spellings

Compatibility aliases read older documents. New manual, benchmark, test, and blueprint sources use canonical spellings. This registry-generated table is the exhaustive alias list; it is empty since the surface swap, when the last four aliases died with the front end that parsed them. `\tenkzkernel` remains callable and rebinds the surface the package already bound at load, so a document that still spells it changes nothing.

Scope	Alias	Replacement
-------	-------	-------------

Aliases normalize to the same semantic events as their replacements and may not change type, default, scope, or diagnostics. The canonical-only linter is authoritative for benchmark cases: replace an alias instead of adding a manifest exception. Historical records in `docs/tenkz/history/` do not define supported aliases.

There is no compatibility form of `\tndefine`: it was an aspirational promise, not a shipped public grammar. Typed custom atoms use `\tndeclareatom`.

THE TENKZ LANGUAGE AT A GLANCE

Choose one layout

tenkz — one typed tensor network
in one frame

Five word classes

atoms — `tn`, `tnfuse`, `tntree`

connections — `tnwire`, `tnbond`,
implicit bonds, cups, traces

annotations — `tnmark`, labels

composition — TeX equations,
alignment, `tngroup`

setup — `tnset`, `tndeclare`, themes

Boundary facts

boundary=open — mints the side legs
and puts them in the signature; an
unstated side draws no leg

boundary=none — the explicit
spelling of the default

boundary=periodic — returns each
row to itself

west=cup / east=cup — connects
adjacent layers, and outranks
`boundary=` on its own side

.tnlog — records the resolved
boundary signature

Canonical habits

one layout — choose from
topology, not silhouette

explicit scope — document,
picture, object, connection

typed ports — virtual joins virtual;
physical joins physical

role/species — semantic style,
never topology

route=arc — the canonical curved
spelling; the family is straight,
`orth`, `arc`

Verification

xelatex — compile the exact source

tenkz_audit.py — audit the
versioned semantic event stream

tenkz_lint.py — reject
non-canonical source

pdftocairo — render and inspect
the affected pages

tenkz_rmp.sh — build the separate
130-target benchmark book